Online Examination Report

Date:	28.07.2014
Attempts:	1
Examinee:	Othieno, Leonard
Date of birth:	10/22/1985
Subject:	080- Principles of flight
Result:	0 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	184	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
2	200	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
3	189	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
4	241	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
5	229	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
6	232	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
7	214	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
8	147	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
9	223	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
10	58	81.1.1.1	Laws and definitions	0,00	1,00
11	37	81.1.1.3	Aerodynamic forces and moments on aerofoils	0,00	1,00
12	30	81.1.2	Two-dimensional airflow around an aerofoil	0,00	1,00
13	95	81.1.2.2	Stagnation point	0,00	1,00
14	23	81.1.2.4	Centre of pressure and aerodynamic centre	0,00	1,00
15	57	81.1.4.2	Induced drag	0,00	1,00
16	93	81.1.4.2	Induced drag	0,00	1,00
17	12	81.1.8.2	The stall speed	0,00	1,00
18	78	81.1.12.1	Ice and other contaminants	0,00	1,00
19	91	81.2.1.3	Influence of temp and alt on M	0,00	1,00
20	16	81.2.2.2	Oblique shock waves	0,00	1,00
21	15	81.4.4.4	Factors affecting static directional stability	0,00	1,00
22	5	81.4.5.5	Factors affecting static lateral stability	0,00	1,00
23	99	81.4.6.3	Dutch roll	0,00	1,00
24	63	81.4.6.3	Dutch roll	0,00	1,00
25	21	81.5.2.4	Location of centre of gravity	0,00	1,00
26	72	81.5.4.5	Means to avoid adverse yaw:	0,00	1,00
27	77	81.5.8.3	Stabiliser trim	0,00	1,00
28	96	81.6.1.1	Flutter	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	6	81.6.1.1	Flutter	0,00	1,00
30	76	81.7.1.1	Relevant propeller parameters	0,00	1,00
31	50	81.7.1.3	Fixed and variable pitch/constant speed	0,00	1,00
32	51	81.7.1.4	Propeller efficiency versus speed	0,00	1,00
33	43	81.7.1.4	Propeller efficiency versus speed	0,00	1,00
34	73	81.7.1.4	Propeller efficiency versus speed	0,00	1,00
35	64	81.7.4.1	Torque reaction	0,00	1,00
36	81	81.7.4.4	Asymmetric blade effect	0,00	1,00
37	1	81.7.4.4	Asymmetric blade effect	0,00	1,00
38	31	81.8.1.4	Straight steady glide	0,00	1,00
39	29	81.8.1.4	Straight steady glide	0,00	1,00
40	68	81.8.1.5	Steady coordinated turn	0,00	1,00
			Total:0%	**0,00**	**40,00**

Attachment 2: List of result by topic

Licence
Date: 28.07.2014
Examinee: Othieno, Leonard
Location: Uganda Civil Aviation Authority

Your Result: 0,00 from 40,00 Points (0,00 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	0,00	40,00	0,00
81.1. SUBSONIC AERODYNAMICS	9	0,00	9,00	0,00
81.2. HIGH SPEED AERODYNAMICS	2	0,00	2,00	0,00
81.4. STABILITY	4	0,00	4,00	0,00
81.5. CONTROL	3	0,00	3,00	0,00
81.6. LIMITATIONS	2	0,00	2,00	0,00
81.7. PROPELLERS	8	0,00	8,00	0,00
81.8. FLIGHT MECHANICS	3	0,00	3,00	0,00
Total	**40**	**0,00**	**40,00**	**0,00**

1

If the nosewheel of an airplane moves aft during gear retraction, how would this aft movement affect the CG location of that airplane? It would

- ☐ have no effect on the CG location.
- ☐ cause the CG location to move forward.
- ☐ cause the CG location to move aft.

ID:	184(LTS:226366, LTM:184)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

2

In comparison to an approach in a moderate headwind, which is an indication of a possible wind shear due to a decreasing headwind when descending on the glide slope?

- ☐ **Higher pitch attitude is required.**
- ☐ **Lower descent rate is required.**
- ☐ **Less power is required.**

ID:	200(LTS:226381, LTM:200)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

3

When one engine fails on a twin-engine airplane, the resulting performance loss

- [ ] reduces cruise indicated airspeed by 50 percent or more.
- [ ] is approximately 50 percent since 50 percent of the normally available thrust is lost.
- [ ] may reduce the rate of climb by 80 percent or more.

ID: 189(LTS:226371, LTM:189)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE

Max. Points 1,00
Points 0,00

4

(Refer to Figure 147.) Which is the correct sequence for recovery from the unusual attitude indicated?

See Attachments:

1. instrument_147.jpg	figure 147	(Attachment No.1)

- ☐ **Add power, lower nose, level wings, return to original attitude and heading.**

- ☐ **Stop turn by raising right wing and add power at the same time, lower the nose, and return to original attitude and heading.**

- ☐ **Level wings, add power, lower nose, descend to original attitude, and heading.**

ID:	241(LTS:226421, LTM:241)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

5

How does the wing design feature 'washout' affect the production of lift?

- ☐ The wing tips continue producing lift when the main body of the wing is not producing lift.
- ☐ The center of lift moves from the trailing edge of the wing, to the leading edge of the wing, as the wing begins to stall.
- ☐ The main body of the wing continues to produce lift when the wing tips are not producing lift.

ID:	229(LTS:226409, LTM:229)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

6

During a 'no-gyro' approach and prior to being handed off to the final approach controller, the pilot should make all turns

- ☐ **one-half standard rate unless otherwise advised.**
- ☐ **standard rate unless otherwise advised.**
- ☐ **any rate not exceeding a 30 degrees bank.**

ID: 232(LTS:226412, LTM:232)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE

Max. Points 1,00
Points 0,00

7

What is the suggested speed to fly when passing through lift with no intention to work the lift?

- ☐ **Minimum sink speed.**
- ☐ **Best glide speed.**
- ☐ **Best lift/drag speed.**

ID:	214(LTS:226395, LTM:214)	Max. Points	1,00
Syllabus reference:	81. PRINCIPLES OF FLIGHT AEROPLANE	Points	0,00

8

When a slight upward or negative flap deflection is used, the result is

- ☐ increased drag.
- ☐ decreased lift.
- ☐ decreased drag.

ID: 147(LTS:226359, LTM:147)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE

Max. Points 1,00
Points 0,00

9

Which is the correct symbol for the stalling speed or the minimum steady flight speed at which the airplane is controllable?

- ☐ **V(S0).**
- ☐ **V(S1).**
- ☐ **V(S).**

ID: 223(LTS:226404, LTM:223)
Syllabus reference: 81. PRINCIPLES OF FLIGHT AEROPLANE

Max. Points 1,00
Points 0,00

10

Considering subsonic incompressible airflow through a Venturi, which statement is correct?

I. The dynamic pressure in the throat is lower than in the undisturbed airflow.
II. The total pressure in the throat is the same as in the undisturbed airflow.

- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is correct.

ID:	58(LTS:226270, LTM:58)	Max. Points	1,00
Syllabus reference:	81.1.1.1 Laws and definitions	Points	0,00

11

The angle between the direction of the undisturbed airflow (relative wind) and the chord line of an aerofoil is called:

- ☐ glide path angle.
- ☐ same as the angle between chord line and fuselage axis.
- ☐ angle of attack.
- ☐ climb path angle.

ID: 37(LTS:226249, LTM:37)
Syllabus reference: 81.1.1.3 Aerodynamic forces and moments on aerofoils

Max. Points 1,00
Points 0,00

See Attachments:

1. 081-0005.gif	Annex	(Attachment No.2)

12

(Please view annex 081-0005 issue date July 2004)
How are the speeds at point 1 and point 2 related in the figure to the relative wind/airflow V?

- ☐ **V1 < V2 and V2 < V.**
- ☐ **V1 > V2 and V2 < V.**
- ☐ **V1 = 0 and V2 = V.**
- ☐ **V1 = 0 and V2 > V.**

ID:	30(LTS:226242, LTM:30)	Max. Points	1,00
Syllabus reference:	81.1.2 Two-dimensional airflow around an aerofoil	Points	0,00

13

Assuming no flow separation, which of these statements about the flow around an aerofoil as the angle of attack increases are correct or incorrect?

I. The stagnation point moves down.
II. The point of lowest static pressure moves aft.

- ☐ I is correct, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.

ID:	95(LTS:226307, LTM:95)	Max. Points	1,00
Syllabus reference:	81.1.2.2 Stagnation point	Points	0,00

14

Assuming no flow separation and no compressibility effects the location of the centre of pressure of a symmetrical aerofoil section:

- ☐ **moves forward when the angle of attack decreases.**
- ☐ **moves backward when the angle of attack decreases.**
- ☐ **is at approximately 25% chord irrespective of angle of attack.**
- ☐ **is at approximately 50% chord irrespective of angle of attack.**

ID:	23(LTS:226234, LTM:23)	Max. Points	1,00
Syllabus reference:	81.1.2.4 Centre of pressure and aerodynamic centre	Points	0,00

15

The induced drag:

- ☐ increases as the aspect ratio increases.
- ☐ increases as the magnitude of the tip vortices decreases.
- ☐ increases as the lift coefficient increases.
- ☐ has no relation to the lift coefficient.

ID:	57(LTS:226269, LTM:57)	Max. Points	1,00
Syllabus reference:	81.1.4.2 Induced drag	Points	0,00

16

Which of these statements about induced drag are correct or incorrect?

I. An elliptical spanwise lift distribution generates more induced drag than a rectangular lift distribution.
II. Induced drag decreases with decreasing aspect ratio.

- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is incorrect.

ID:	93(LTS:226305, LTM:93)	Max. Points	1,00
Syllabus reference:	81.1.4.2 Induced drag	Points	0,00

17

The stall speed:

- ☐ decreases with an increased weight.
- ☐ increases with the length of the wingspan.
- ☐ increases with an increased weight.
- ☐ does not depend on weight.

ID: 12(LTS:226224, LTM:12)
Syllabus reference: 81.1.8.2 The stall speed

Max. Points 1,00
Points 0,00

18

The effects of very heavy rain (tropical rain) on the aerodynamic characteristics of an aeroplane are:

- ☐ increase of CLMAX and increase of drag.
- ☐ decrease of CLMAX and decrease of drag.
- ☐ decrease of CLMAX and increase of drag.
- ☐ increase of CLMAX and decrease of drag.

ID:	78(LTS:226290, LTM:78)	Max. Points	1,00
Syllabus reference:	81.1.12.1 Ice and other contaminants	Points	0,00

19

If the altitude is increased and the TAS remains constant in the troposphere under standard atmospheric conditions, the Mach number will:

- ☐ increase or decrease, depending on the type of aeroplane.
- ☐ increase.
- ☐ not change.
- ☐ decrease.

ID:	91(LTS:226303, LTM:91)	Max. Points	1,00
Syllabus reference:	81.2.1.3 Influence of temp and alt on M	Points	0,00

20

Compared with an oblique shock wave a normal shock wave has a:

- ☐ lower static temperature.
- ☐ higher loss in total pressure.
- ☐ higher total temperature.
- ☐ higher total pressure.

ID:	16(LTS:226228, LTM:16)	Max. Points	1,00
Syllabus reference:	81.2.2.2 Oblique shock waves	Points	0,00

21

Which of the following provides a positive contribution to static directional stability?

- ☐ A forward swept wing.
- ☐ A low wing as compared with a high wing.
- ☐ Moving the centre of gravity aft.
- ☐ A dorsal fin.

ID:	15(LTS:226227, LTM:15)	Max. Points	1,00
Syllabus reference:	81.4.4.4 Factors affecting static directional stability	Points	0,00

22

Sweepback of a wing positively influences:

1. static longitudinal stability.
2. static lateral stability.
3. dynamic longitudinal stability.

The combination that regroups all of the correct statements is:

- ☐ **3.**
- ☐ **1.**
- ☐ **1, 2, 3.**
- ☐ **2.**

ID:	5(LTS:226217, LTM:5)	Max. Points	1,00
Syllabus reference:	81.4.5.5 Factors affecting static lateral stability	Points	0,00

23

An aeroplane is sensitive to Dutch roll when:

- [ ] static lateral stability is much more pronounced than static directional stability.
- [ ] an aeroplane has anhedral.
- [ ] static lateral stability is much less pronounced than static directional stability.
- [ ] static stability is less pronounced than dynamic stability.

ID:	99(LTS:226311, LTM:99)	Max. Points	1,00
Syllabus reference:	81.4.6.3 Dutch roll	Points	0,00

24

Which aeroplane behaviour will be corrected by a yaw damper?

- ☐ **Spiral dive.**
- ☐ **Flutter.**
- ☐ **Dutch roll.**
- ☐ **Buffeting.**

ID:	63(LTS:226275, LTM:63)	Max. Points	1,00
Syllabus reference:	81.4.6.3 Dutch roll	Points	0,00

25

Which statement is about CG limits is correct?

- ☐ The aft CG limit is determined by the maximum elevator deflection available.
- ☐ The forward CG limit is mainly determined by the amount of pitch control available from the elevator.
- ☐ If the aft CG limit is correctly chosen, the forward CG limit is automatically determined as well.
- ☐ The forward CG limit is determined by stability considerations only.

ID:	21(LTS:226233, LTM:21)	Max. Points	1,00
Syllabus reference:	81.5.2.4 Location of centre of gravity	Points	0,00

26

An example of differential aileron deflection during initiation of left turn is:

- ☐ **Left aileron: 2° up.**
  Right aileron: 5° down.
- ☐ **Left aileron: 2° down.**
  Right aileron: 5° up.
- ☐ **Left aileron: 5° up.**
  Right aileron: 2° down.
- ☐ **Left aileron: 5° down.**
  Right aileron: 2° up.

ID:	72(LTS:226284, LTM:72)	Max. Points	1,00
Syllabus reference:	81.5.4.5 Means to avoid adverse yaw:	Points	0,00

27

The most important factor determining the required position of the Trimmable Horizontal Stabiliser (THS) for take off is the:

- ☐ total mass of the aeroplane.
- ☐ stall speed.
- ☐ position of the aeroplane's centre of gravity.
- ☐ centre of gravity position of the fuel.

ID:	77(LTS:226289, LTM:77)	Max. Points	1,00
Syllabus reference:	81.5.8.3 Stabiliser trim	Points	0,00

28

Wing flutter may be caused by a:

- ☐ **aerodynamic wing stall at high speed.**
- ☐ **combination of bending and torsion of the wing structure.**
- ☐ **combination roll control reversal and low speed stall.**
- ☐ **combination of fuselage bending and wing torsion.**

ID:	96(LTS:226308, LTM:96)	Max. Points	1,00
Syllabus reference:	81.6.1.1 Flutter	Points	0,00

29

Which of these statements about flutter are correct or incorrect?

I. If flutter occurs, IAS should be reduced.
II. Resistance to flutter increases with increasing wing stiffness.

- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is incorrect.

ID: 6(LTS:226218, LTM:6)
Syllabus reference: 81.6.1.1 Flutter

Max. Points 1,00
Points 0,00

30

The difference between a propeller's blade angle and its angle of attack is called:

- ☐ **the propeller angle.**
- ☐ **the helix angle.**
- ☐ **propeller slip.**
- ☐ **the effective pitch.**

ID: 76(LTS:226288, LTM:76)
Syllabus reference: 81.7.1.1 Relevant propeller parameters

Max. Points 1,00
Points 0,00

31

The angle of attack of a fixed pitch propeller blade increases when:

- ☐ velocity and RPM increase.
- ☐ velocity and RPM decrease.
- ☐ forward velocity increases and RPM decreases.
- ☐ RPM increases and forward velocity decreases.

ID:	50(LTS:226262, LTM:50)	Max. Points	1,00
Syllabus reference:	81.7.1.3 Fixed and variable pitch/constant speed	Points	0,00

32

Which statement is correct when comparing a fixed pitch propeller with a constant speed propeller?

I. A fixed pitch propeller improves propeller efficiency over a range of cruise speeds.
II. A coarse fixed pitch propeller is more efficient during take-off.

- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is incorrect.

ID:	51(LTS:226263, LTM:51)	Max. Points	1,00
Syllabus reference:	81.7.1.4 Propeller efficiency versus speed	Points	0,00

33

(Please view annex 081-0037 issue date November 2005)
A typical curve representing propeller efficiency of a fixed pitch propeller versus TAS at constant RPM is:

See Attachments:

1. 081-0037.gif Annex (Attachment No.3)

- ☐ **diagram 2.**
- ☐ **diagram 4.**
- ☐ **diagram 1.**
- ☐ **diagram 3.**

ID: 43(LTS:226254, LTM:43)
Syllabus reference: 81.7.1.4 Propeller efficiency versus speed

Max. Points 1,00
Points 0,00

34

Which statement is correct when comparing a fixed pitch propeller with a constant speed propeller?

I. A constant speed propeller reduces fuel consumption over a range of cruise speeds.
II. A coarse fixed pitch propeller is more efficient during take-off.

- ☐ **I is incorrect, II is correct.**
- ☐ **I is incorrect, II is incorrect.**
- ☐ **I is correct, II is incorrect**
- ☐ **I is correct, II is correct.**

ID:	73(LTS:226285, LTM:73)	Max. Points	1,00
Syllabus reference:	81.7.1.4 Propeller efficiency versus speed	Points	0,00

35

The torque reaction of a rotating fixed pitch propeller will be greatest at:

- ☐ **low aeroplane speed and maximum engine power.**
- ☐ **high aeroplane speed and low engine power.**
- ☐ **low aeroplane speed and low engine power.**
- ☐ **high aeroplane speed and maximum engine power.**

ID:	64(LTS:226276, LTM:64)	Max. Points	1,00
Syllabus reference:	81.7.4.1 Torque reaction	Points	0,00

36

A propeller is turning to the right when viewed from behind; The asymmetric blade effect in the climb at low speed will:

- ☐ roll the aeroplane to the right.
- ☐ yaw the aeroplane to the left.
- ☐ yaw the aeroplane to the right.
- ☐ roll the aeroplane to the left.

ID:	81(LTS:226293, LTM:81)	Max. Points	1,00
Syllabus reference:	81.7.4.4 Asymmetric blade effect	Points	0,00

37

Which statement about a propeller is correct?

I. Asymmetric blade effect is unaffected when engine power is increased.
II. Asymmetric blade effect is independent of the angle between the propeller axis and the
airflow through the propeller disc.

- ☐ **I is correct, II is incorrect.**
- ☐ **I is incorrect, II is incorrect.**
- ☐ **I is correct, II is correct.**
- ☐ **I is incorrect, II is correct.**

ID:	1(LTS:226213, LTM:1)	Max. Points	1,00
Syllabus reference:	81.7.4.4 Asymmetric blade effect	Points	0,00

38

What factors determine the distance travelled over ground of an aeroplane in a glide from a given altitude?

- ☐ The wind and weight together with power loading, which is the ratio of power output to the weight.
- ☐ The wind and the lift/drag ratio.
- ☐ The wind and CLMAX.
- ☐ The wind and the aeroplane's mass.

ID:	31(LTS:226243, LTM:31)	Max. Points	1,00
Syllabus reference:	81.8.1.4 Straight steady glide	Points	0,00

39

Which of the following increases the maximum duration of a glide?

- ☐ A decrease in mass.
- ☐ A headwind.
- ☐ A tailwind.
- ☐ An increase in mass.

ID:	29(LTS:226241, LTM:29)	Max. Points	1,00
Syllabus reference:	81.8.1.4 Straight steady glide	Points	0,00

40

An aeroplane performs a steady horizontal, co-ordinated turn with 45 degrees of bank at 230 kt TAS. The same aeroplane with the same bank angle and speed, but at a higher mass:

- ☐ **will turn with a smaller turn radius.**
- ☐ **will turn with a larger turn radius.**
- ☐ **will turn with the same radius, but might stall.**
- ☐ **will have a higher rate of turn.**

ID:	68(LTS:226280, LTM:68)	Max. Points	1,00
Syllabus reference:	81.8.1.5 Steady coordinated turn	Points	0,00

Attachment No. 1

Questions 4
figure 147
instrument_147.jpg

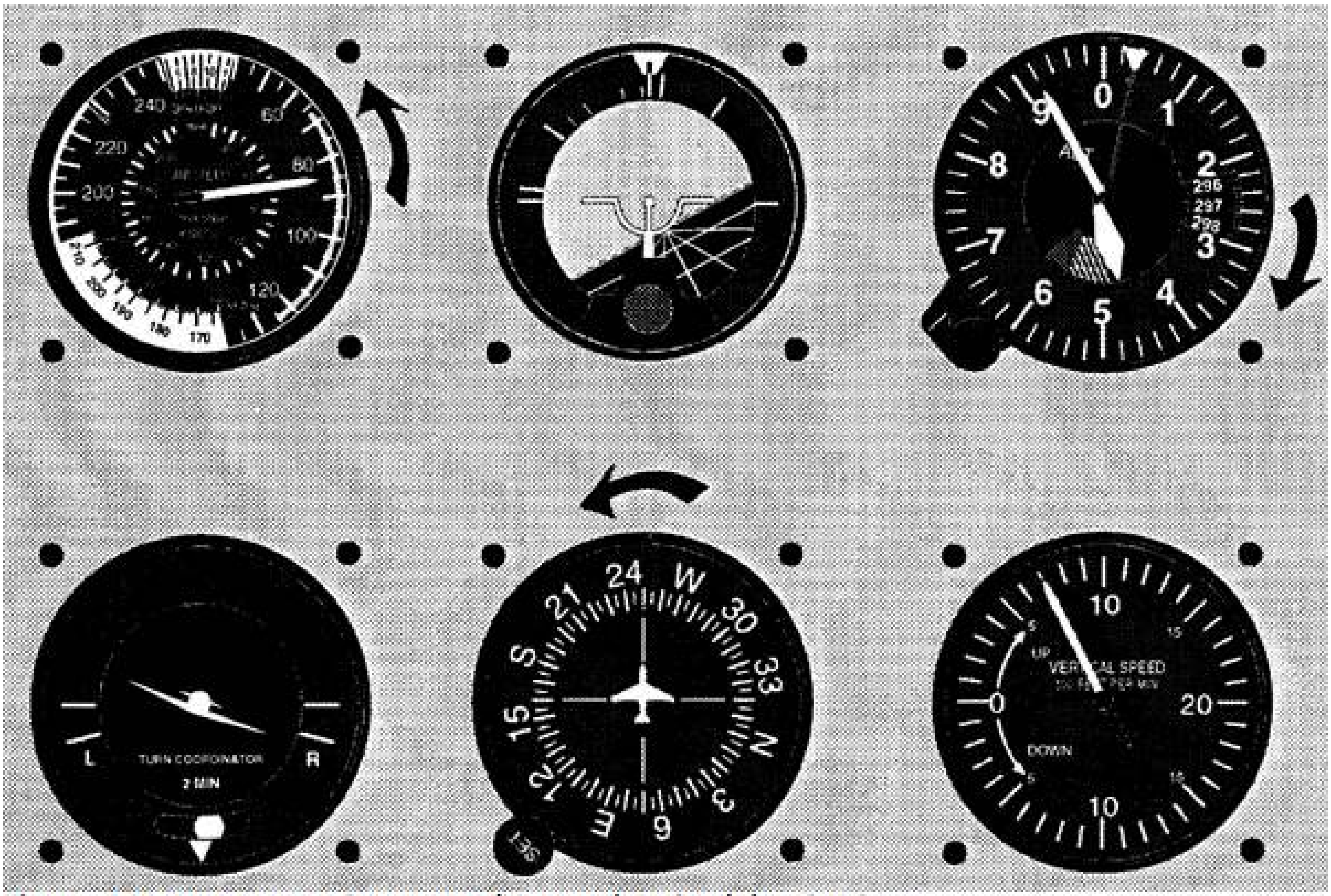


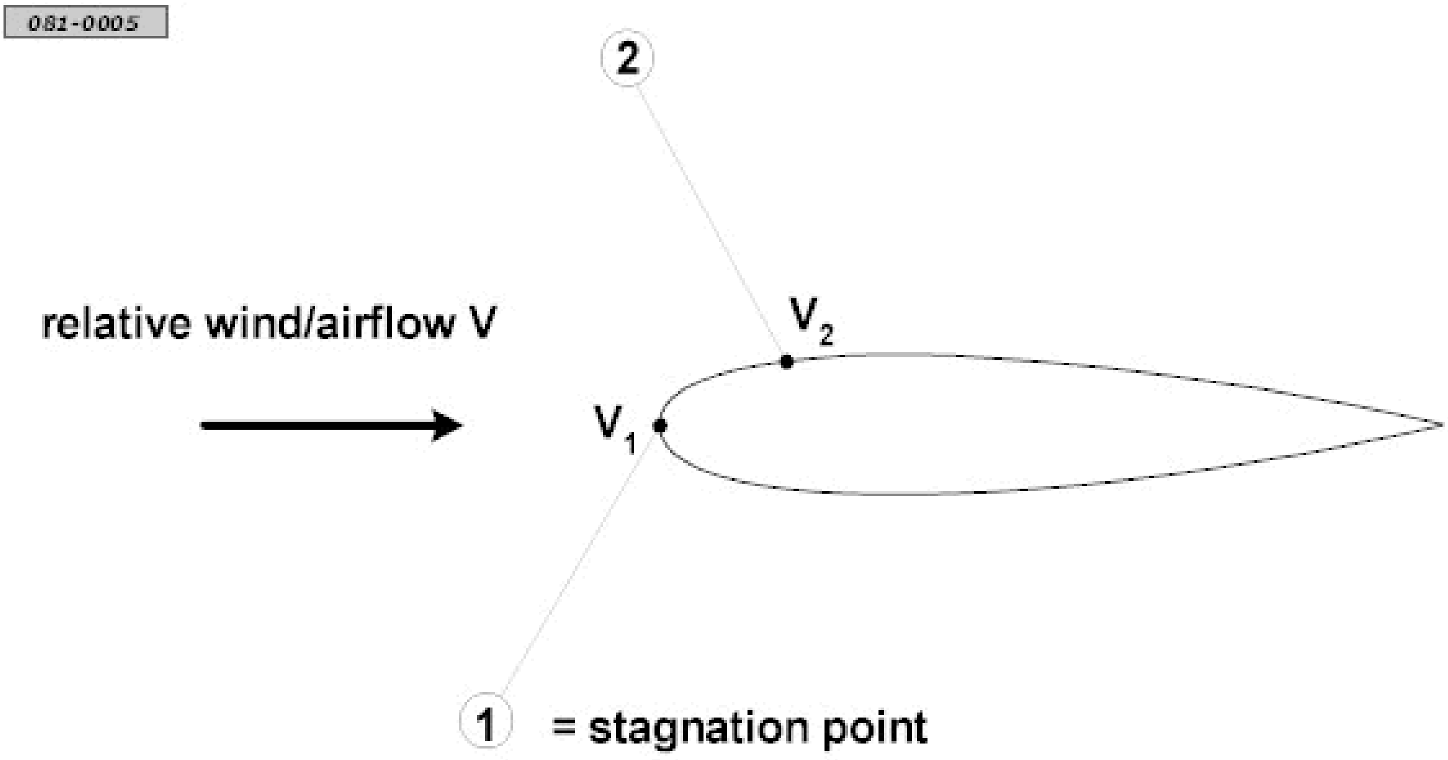
Figure 147. Instrument Sequence (Unusual Attitude) © ASA

Attachment No. 2

Questions 12

Annex

081-0005.gif



Attachment No. 3

Questions 33

Annex

081-0037.gif